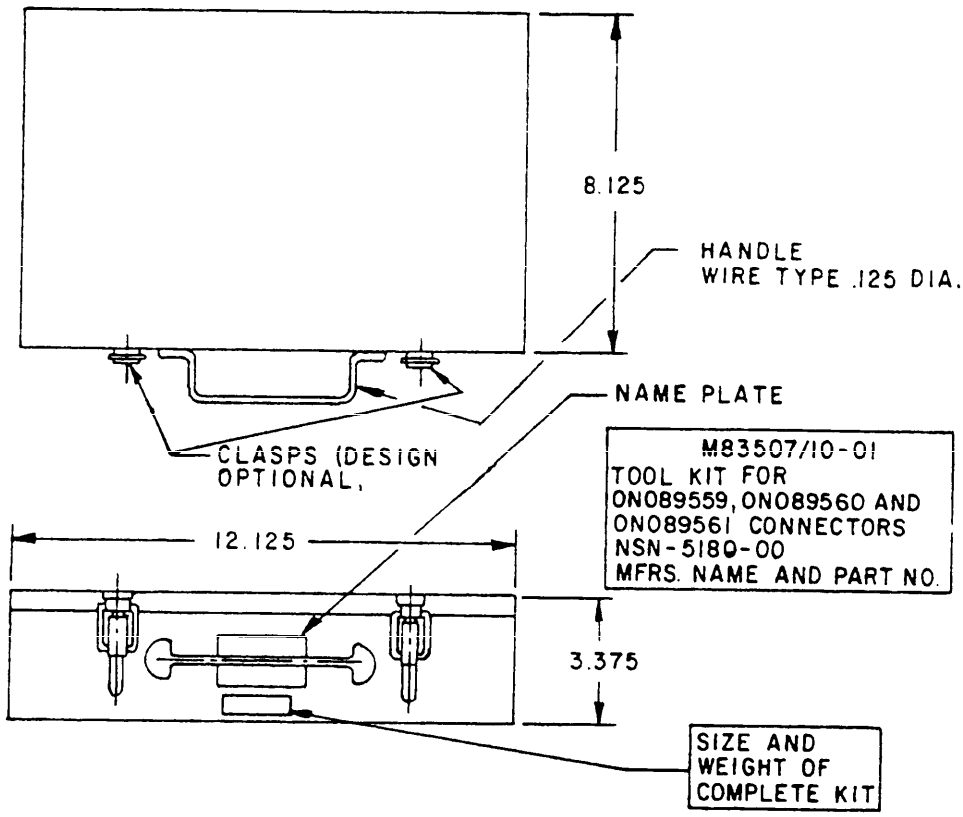
MILITARY SPECIFICATION SHEET

TOOL KITS, ELECTRICAL CONNECTORS, CONTACTS, AND CONNECTOR AND CABLE ACCESSORIES: TOOL KIT FOR ON089559, ON089560, AND ON089561 CONNECTORS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tool kit described herein shall consist of this document and the latest issue of Specification MIL-T-83507.



NOTES:
1. For drawing notes and metrics see figure 5.
2. Part number M83507/10-01 includes case and tools.

FIGURE 1. Tool kit for ON089559, ON089560, and ON089561 connectors.

FSC 5180

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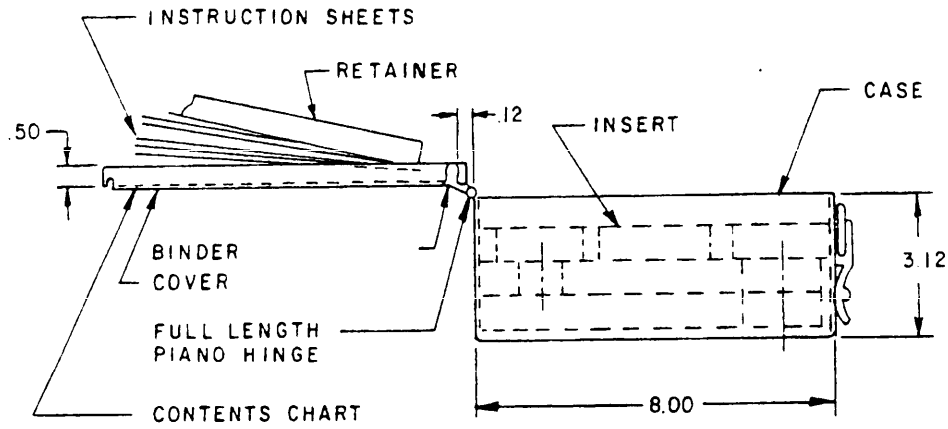
PARTS LIST:

(1)	M83507/10-02	-	Tool case assembly.
(1)	M22520/1-01	-	Crimp tool frame.
(1)	M22520/2-01	-	Crimp tool frame.
(1)	M22520/10-01	-	Crimp tool frame.
(1)	M22520/1-02	-	Turret head.
(1)	M22520/2-24	-	Positioner.
(1)	M22520/10-06	-	Die set.
(1)	M22520/3-1	-	In-service inspection gage.
(1)	9/64 hex wrench.		
(2)	ON089564	-	Installing tool.
(2)	ON089565	-	Removal tool.
(3)	M81969/14-03	-	Installing and removal tool.
(3)	M81969/14-02	-	Installing and removal tool.

(1 set) Contents charts, instruction sheets, and tool selection chart.

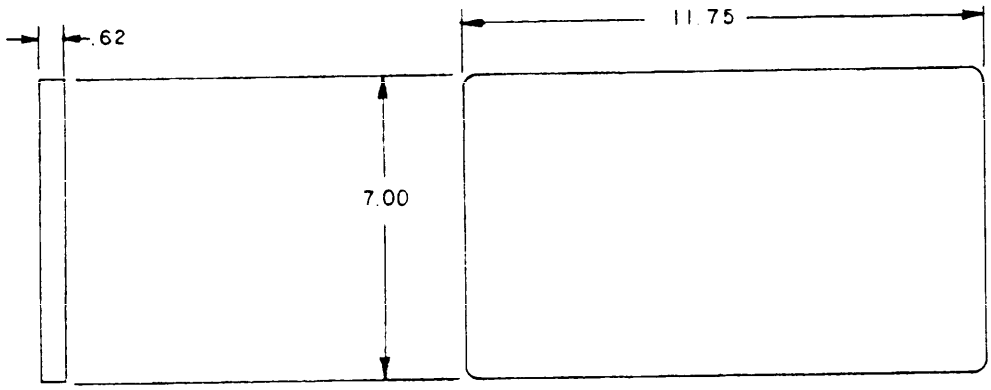
FIGURE 1. Tool kit for ON089559, ON089560, and ON089561 connectors - Continued.

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NOTES:
1. For drawing notes and metrics, see figure 5.
2. Part number M83507/10-02 for case only (tools not included).

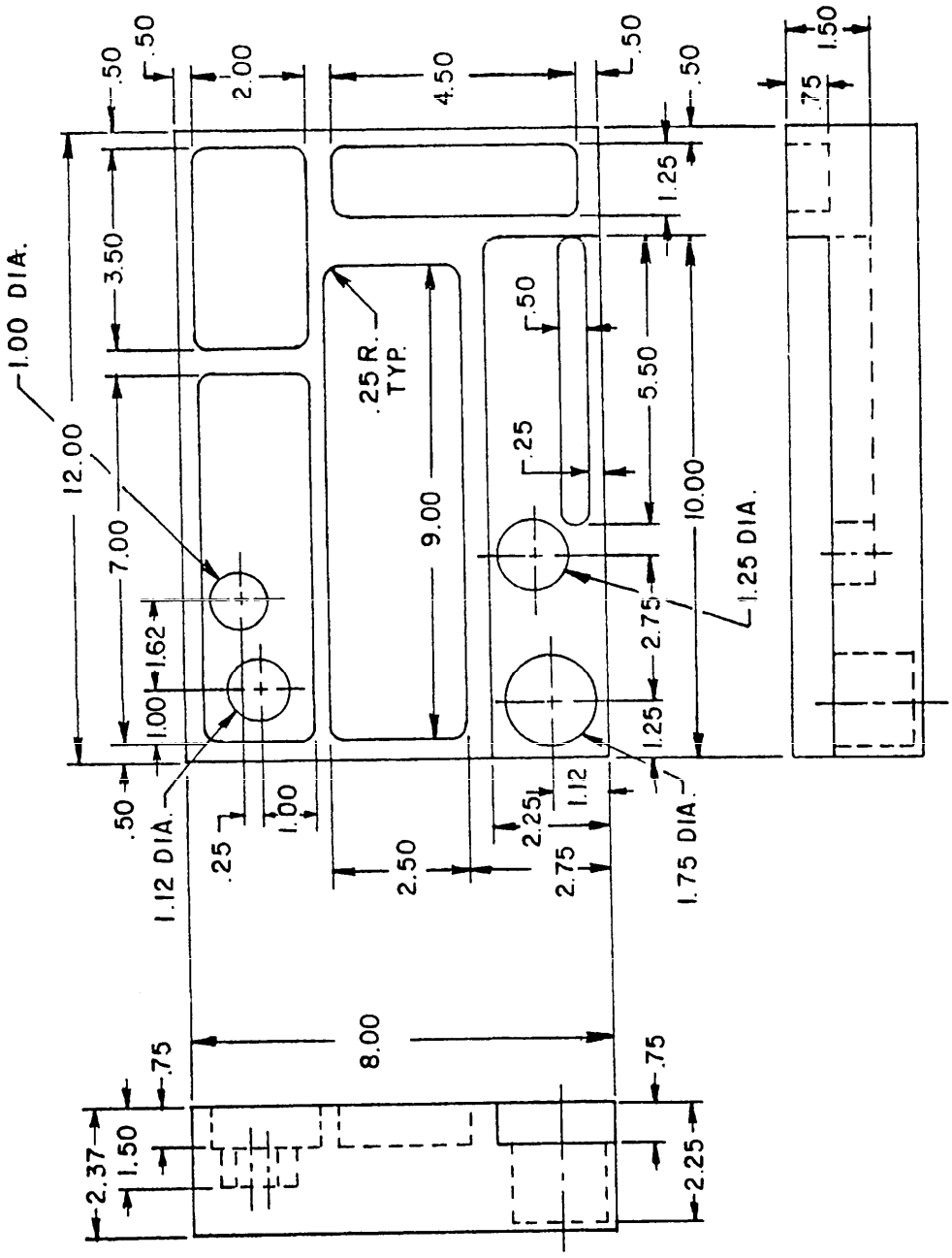
FIGURE 2. Tool case assembly.



NOTES:
1. For drawing notes and metric equivalents, see figure 5.
2. This retainer is part of the tool case assembly (see figure 2).

FIGURE 3. Retainer.

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NOTE:
1. For drawing notes and metric equivalents, see figure 5.
2. This insert is part of the tool case assembly (see figure 2).

FIGURE 4. Insert.

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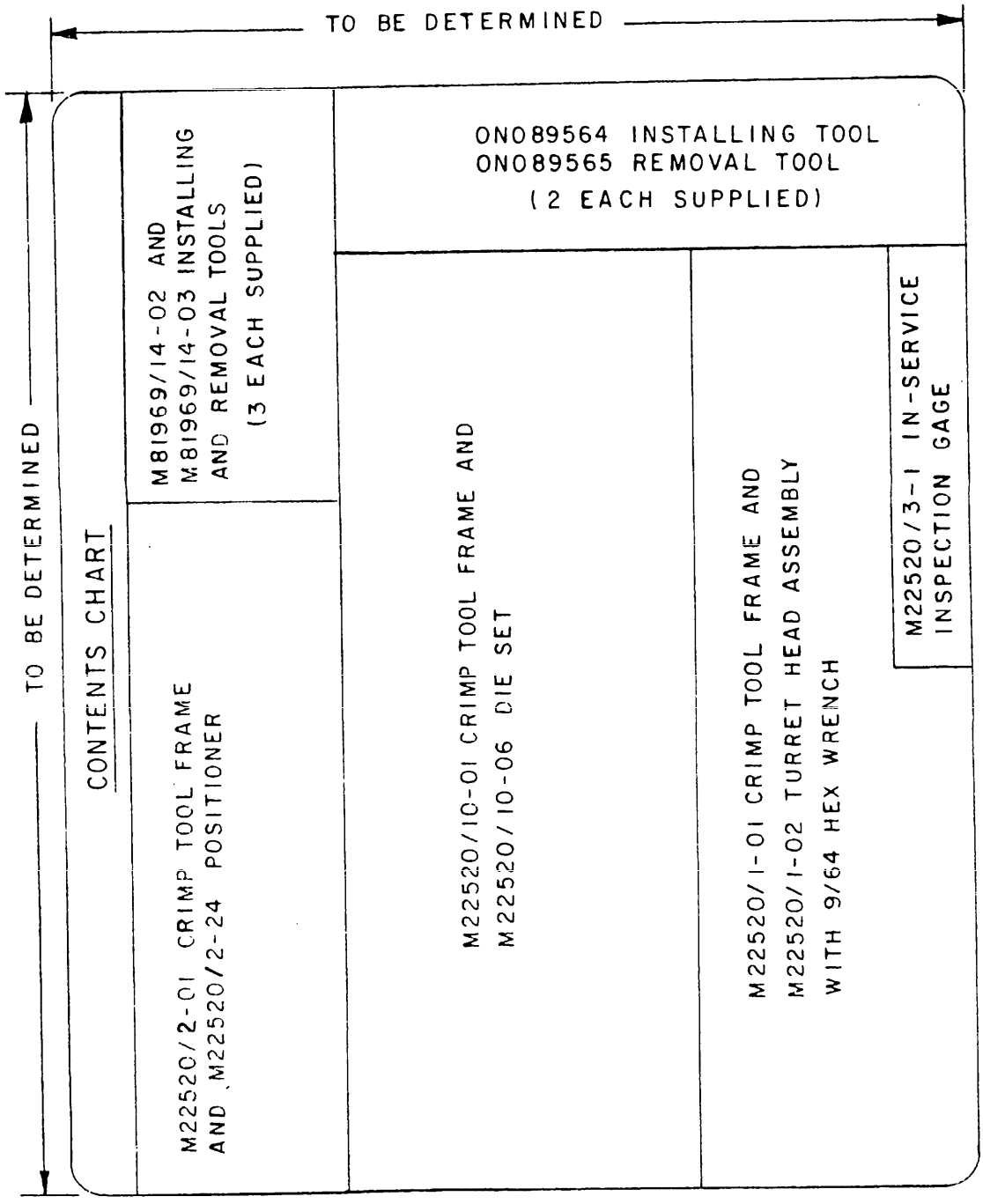
INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.12	3.0	1.25	31.8	2.50	63.5	7.00	177.8
.125	3.18	1.50	38.1	2.75	69.8	8.00	203.2
.25	6.4	1.62	41.1	3.12	79.2	8.125	206.38
.50	12.7	1.75	44.4	3.375	85.72	9.00	228.6
.75	19.0	2.00	50.8	3.50	88.9	10.00	254.0
1.00	25.4	2.25	57.2	4.50	114.3	12.00	304.8
1.12	28.4	2.37	60.2	5.50	139.7	12.125	307.98

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.031 (.79 mm).
4. Remove all burrs, break sharp edges and corners.

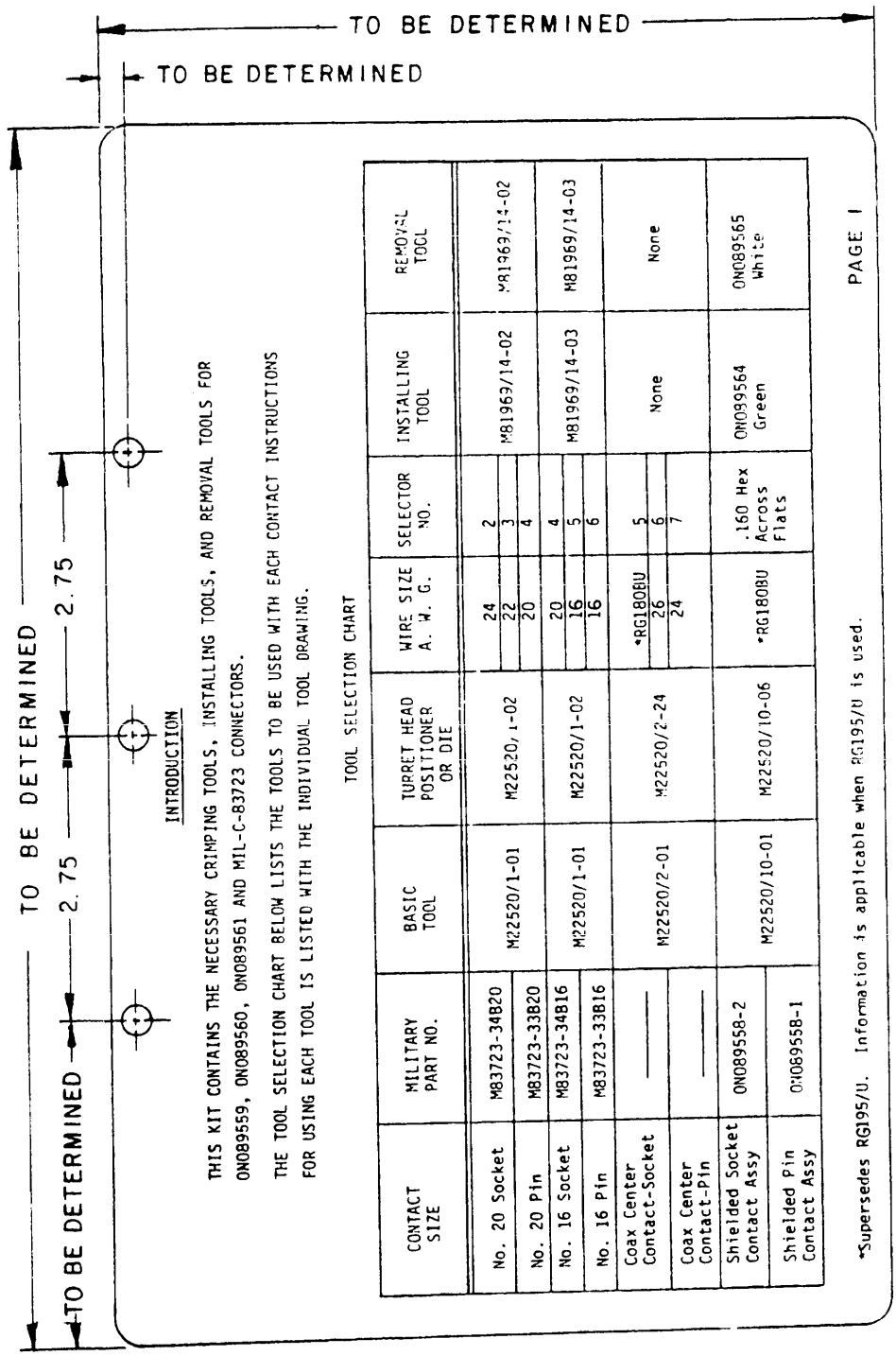
FIGURE 5. Drawing notes and metric equivalents.

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INSTRUCTION SHEETS ARE AS FOLLOWS:

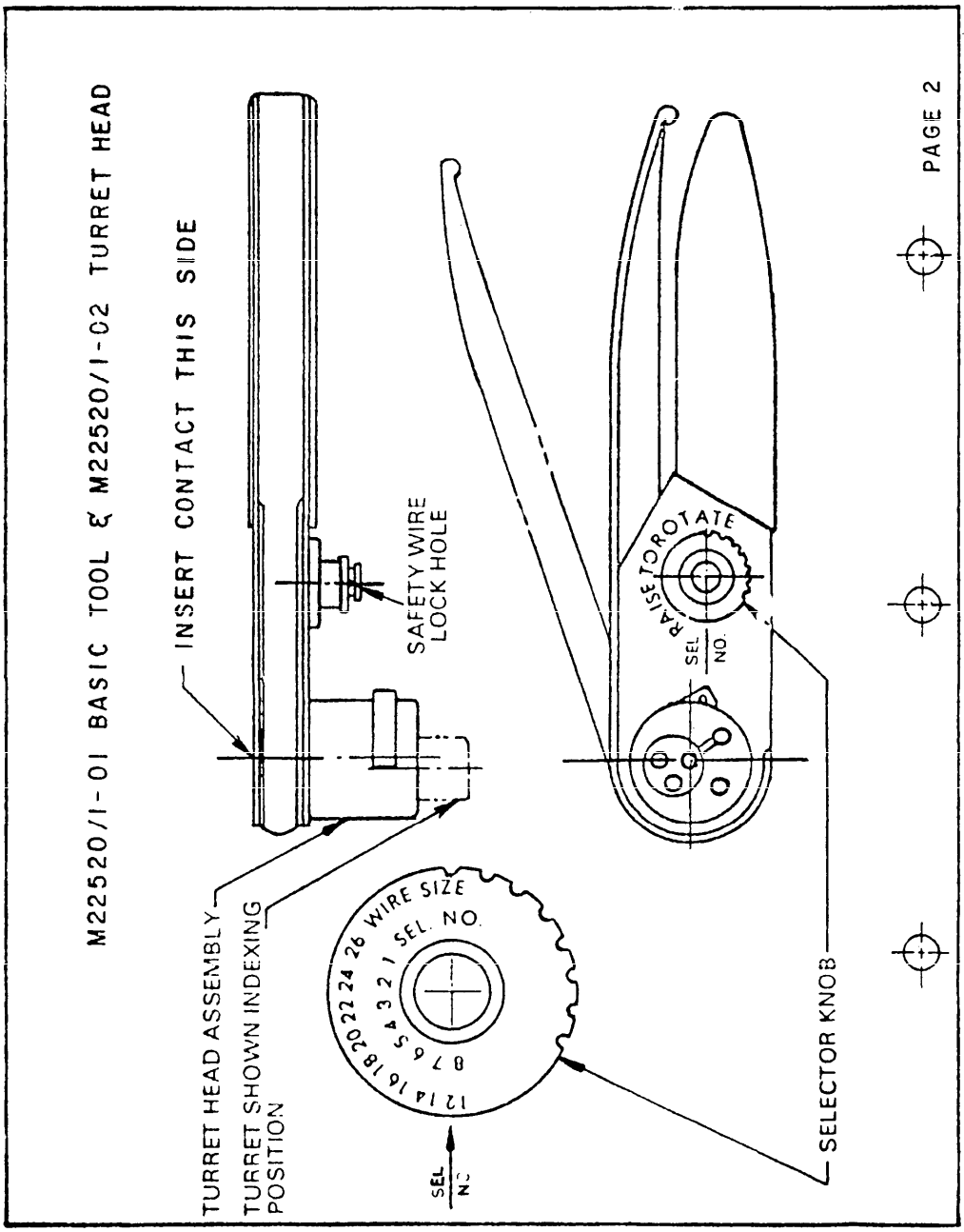


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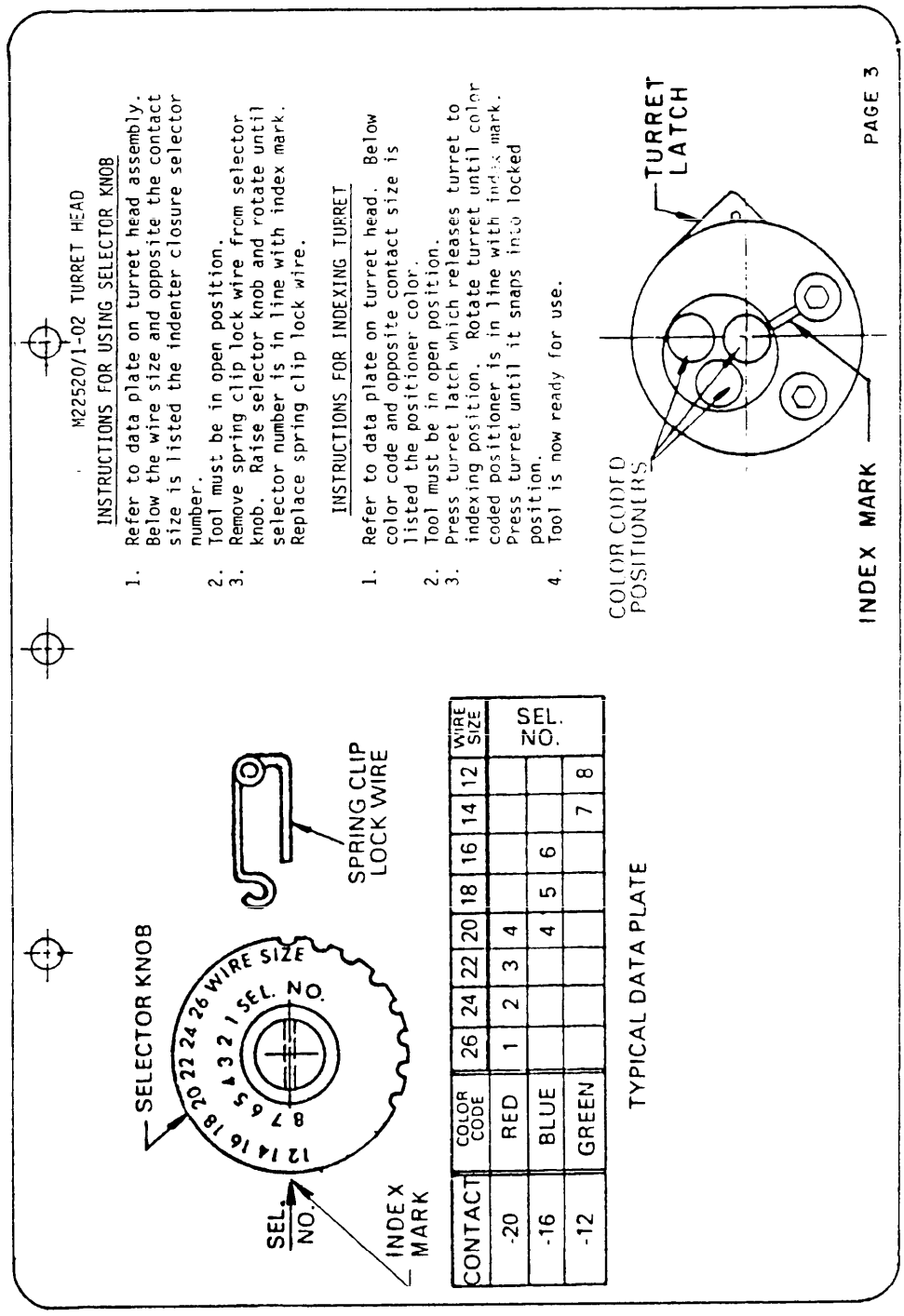


CONTACT SIZE	MILITARY PART NO.	BASIC TOOL	TURRET HEAD POSITIONER OR DIE	WIRE SIZE A. W. G.	SELECTOR NO.	INSTALLING TOOL	REMOVAL TOOL
No. 20 Socket	M83723-34B20	M22520/1-01	M22520/1-02	24	2	M81969/14-02	M81969/14-02
				22	3		
				20	4		
No. 20 Pin	M83723-33B20						
No. 16 Socket	M83723-34B16	M22520/1-01	M22520/1-02	20	4	M81969/14-03	M81969/14-03
				16	5		
				16	6		
No. 16 Pin	M83723-33B16						
Coax Center Contact-Socket	———	M22520/2-01	M22520/2-24	*RG180BU	5	None	None
				26	6		
				24	7		
Coax Center Contact-Pin	———						
Shielded Socket Contact Assy	ON089558-2	M22520/10-01	M22520/10-06	*RG180BU	.160 Hex Across Flats	ON089564 Green	ON089565 White
Shielded Pin Contact Assy	ON089558-1						

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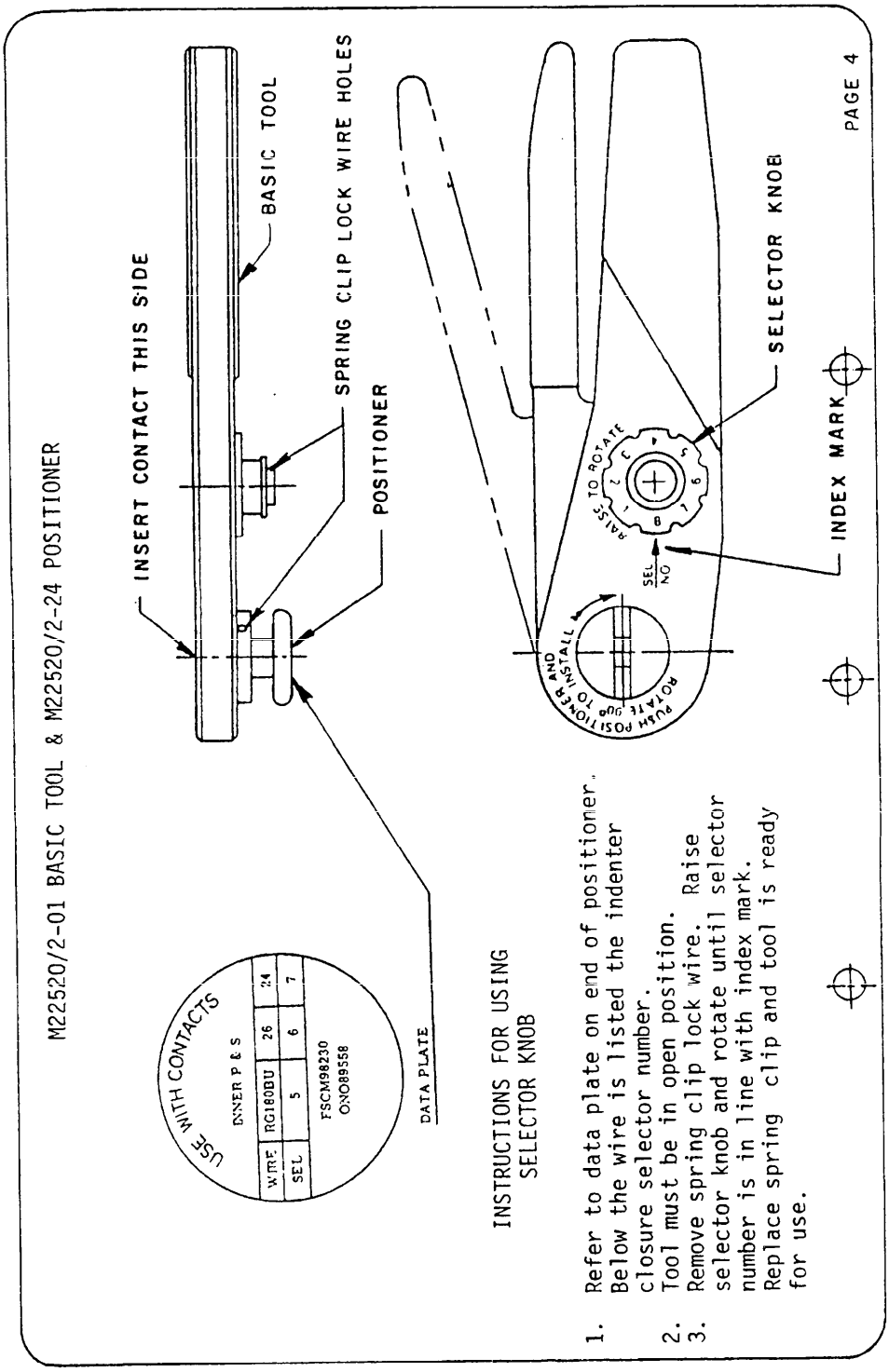


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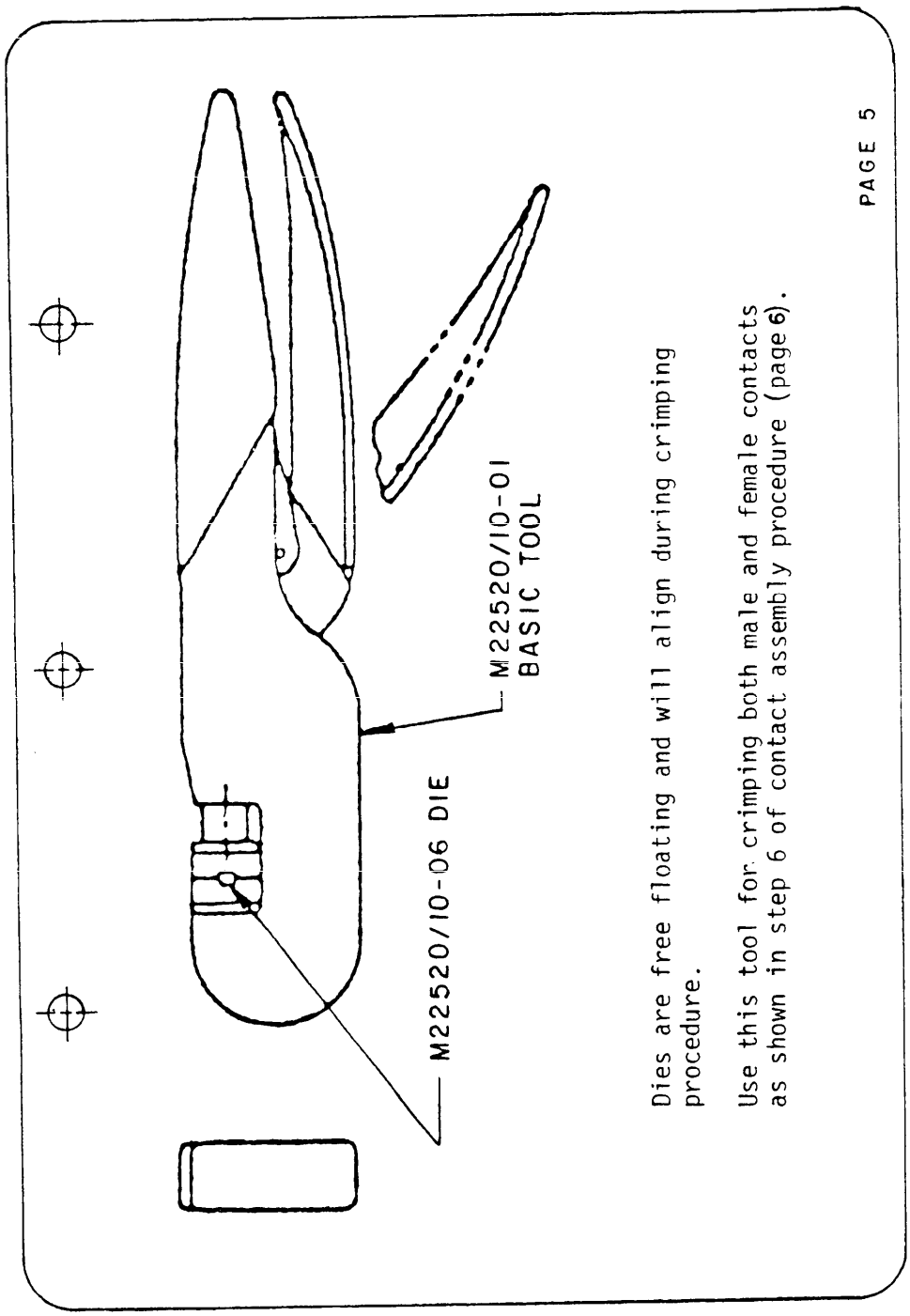


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Strainer basket and bangle. The strainer basket and bangle



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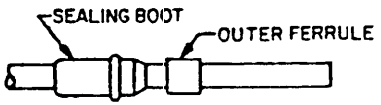


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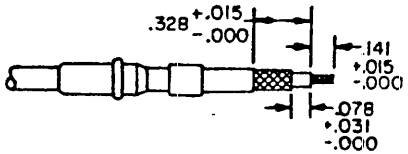
ASSEMBLY INSTRUCTIONS FOR SHIELDED CONTACTS (COAX)

SAME FOR MALE AND FEMALE



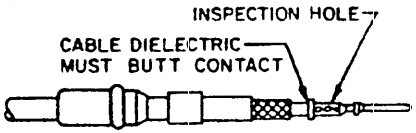
Step 1

Install sealing boot and slide outer ferrule onto cable. As shown, prior to stripping.



Step 2

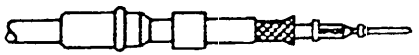
Strip cable jacket. Braid and dielectric to dimensions indicated. All cuts are to be square and sharp. Important: Do not nick braid, dielectric or center conductor when cutting. If wire ends of center conductor fray, twist them to their normal lay.



Step 3

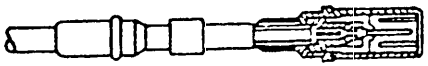
Place stripped center conductor into contact until cable dielectric butts contact and end of wire shows through inspection hole (arrow). Insert contact into crimp tool until fully seated in positioner. Crimp with one full stroke. See chart below for crimp tool setting.

CRIMP ON CENTER CONTACT			
Cable	Tool	Positioner	Setting
RG180BU	M22520/2-01	M22520/2-24	No. 5



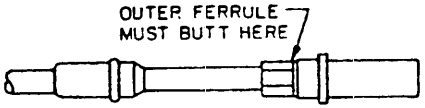
Step 4

Flare out slightly, ends of wire braid to facilitate insertions on inner ferrule of body assembly. (Do not comb out wire braid.)



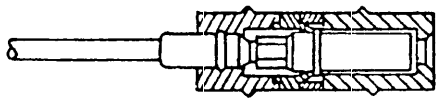
Step 5

Install center contact in body assembly and slide inner ferrule underneath wire braid as shown. Center contact shall be inserted until it is locked in place in body assembly. A 3 lb. min. 5 lb. max. pull shall be exerted on cable to ensure that the contact is securely locked in place.



Step 6

Slide outer ferrule over wire braid and up against body assembly as shown. No slack shall exist in wire braid. Crimp outer ferrule with M22520/10-01 basic tool and M22520/10-06 die.

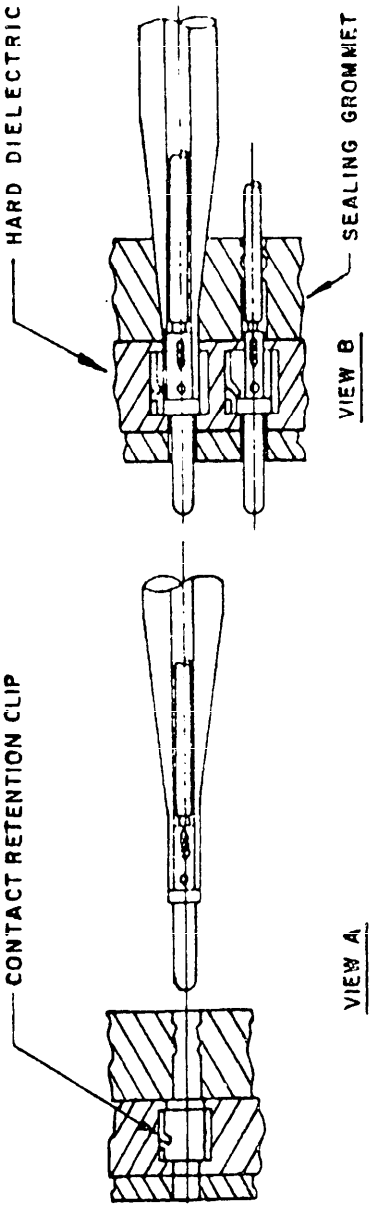


Step 7

The shielded contact now assembled shall be inserted into the connector assembly using ON089564 INSTALLING tool. Push sealing boot forward into grommet of connector until O-ring riser of the boot has effected its snap-in seal. The installation of the shielded contact is then completed for removing contact from connector use ON089565 REMOVAL TOOL.

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INSTRUCTION FOR PLASTIC CONTACT INSTALLING TOOLS



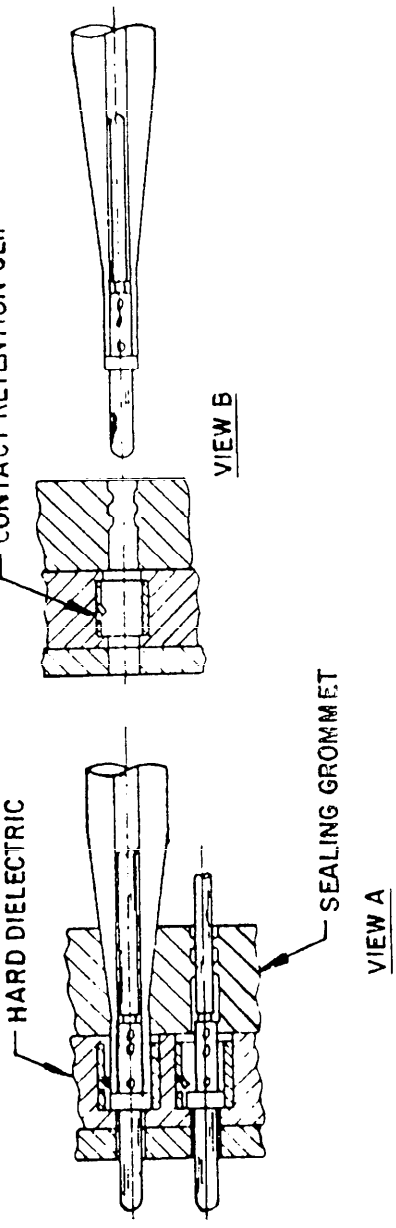
1. Select correct installing tool (See Tool Selection Chart) and place tool around wire attached to contact to be inserted.
2. Slide tool down wire on to contact crimp barrel until it butts against the contact shoulder. (See View A)
3. Grip tool, wire, and contact as a unit and guide contact into connector thru sealing grommet and hard dielectric until contact is firmly seated. (See View B)
4. Release grip on wire and withdraw tool from connector until it clears the grommet. Hold wire forward while tool is being removed from wire.

CAUTION: Assembled contact must be held perpendicular to face of connector during installing. When seating boot do not allow excessive bending or twisting of the cable.

PAGE 7

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INSTRUCTION FOR PLASTIC CONTACT REMOVAL TOOLS

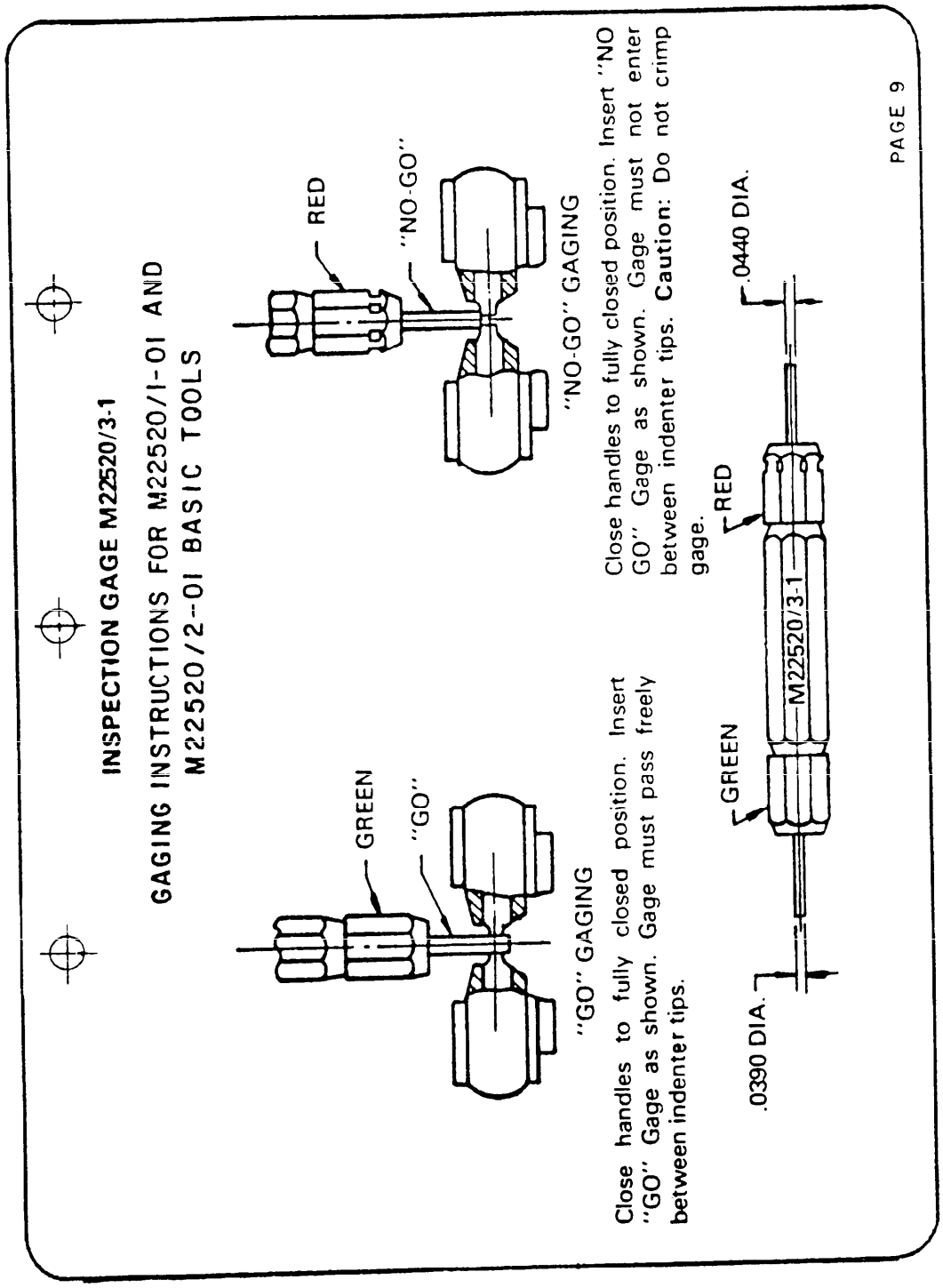


1. Select correct removal tool (See Tool Selection Chart) and place tool around wire attached to contact to be removed.
2. Slide tool down wire into the connector thru the sealing grommet and into the hard dielectric until tool stops against contact shoulder. (See View A)
3. At this stage of removal, tool will collapse contact retention clips to the point where they no longer retain control.
4. Grip wire and tool as a unit and withdraw wire, tool and contact from connector simultaneously. (See View B)

CAUTION: To avoid damage to tool and/or connector, do not tip, spread or rotate tool while in connector.

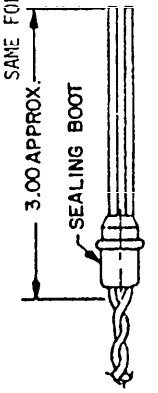
PAGE 8

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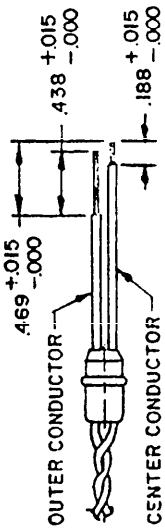
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PROCEDURE FOR CONTACTS (TWISTED PAIR)
SAME FOR MALE AND FEMALE, (MALE NOT SHOWN)



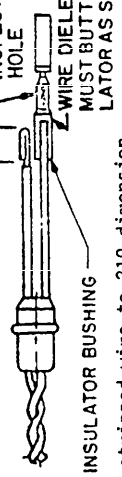
Step 1

Install sealing boot onto the untwisted wire, as shown, prior to stripping wires.



Step 2

Strip wires to dimensions indicated. All cuts are to be square and sharp. Do not nick conductors during stripping of insulation.



Step 3

1. Fold longer stripped wire to .219 dimension shown.
2. Place insulator and contact onto shorter stripped wire as shown.
3. Wire center conductor must be visible thru contact inspection hole (arrow).
4. Set selector number 7 on M22520/2-01 basic tool with M22520/2-24 positioner.
5. Insert contact into crimp tool until fully seated in positioner.
6. Crimp with one full stroke.

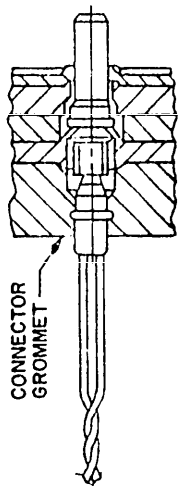


Step 4

Slide outer ferrule onto cable as shown prior to installing center contact in body assembly. Position folded wire on top of inner ferrule as shown. Center contact shall be inserted until it is locked in place in body assembly. A 3 lb. min. 5 lb. max. pull shall be exerted on cable to ensure that the contact is securely locked in place.

Step 5

Slide outer ferrule over folded wire against body assembly as shown. Crimp outer ferrule with M22520/10-01 basic tool and M22520/10-06 die.



Step 6

The shielded contact now assembled shall be inserted into the connector assembly using ONO89564installing tool. Push sealing boot forward into the grommet until O-ring riser of the boot has effected its snap-in seal. The installation of the shielded contact is now completed. For removing contact from connector use ONO89565 removal tool.

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REQUIREMENTS:

Design and configuration: See figures 1 through 5.

Material:
Case and cover (category 3): Sheet metal, 21 gage in accordance with QQ-S-698, gray baked enamel on all metal parts.
Instruction sheets and contents charts: White vinyl 0.020 inch (0.51 mm) thick.
Insert and retainer pad: Fire retardant foamed polyethylene in accordance with PPP-C-1752; color is optional.

Binder: Looseleaf, three ring, 0.562 inch (14.27 mm) ±0.062 inch (1.57 mm) I.D., 2.75 inches (69.9 mm) center distance between rings, securely riveted to inside of cover as shown.

Clasps, draw pull type with provision for locking device, handles, and hinges: Plated to resist corrosion.

Nameplate: In accordance with MIL-P-19834.

Instruction sheets and contents chart (see pages 6 through 16): Lettering shall be legible and protected with a transparent coating. Contents chart (centered on inside of case cover) shall have pressure sensitive adhesive backing and show the insert cavity configurations and the respective tool part numbers; a list of NSN's will also be supplied.

Part number:
M83507/10-01 - Tool kit for ONO89559, ONO89560, and ONO89561 connectors.
M83507/10-02 - Tool case assembly.

QUALIFICATION:

Qualifying activity: Defense Electronics Supply Center (DESC-E),Dayton, OH 45444.

Custodians:
Army - EL
Navy - AS
Air Force - 85

Preparing activity:
Air Force - 85

(Project 5180-0101)

Review activities:
Army - MI, AR, AV, WC, GL
Navy - SH, EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT
Navy - MC
Air Force - 19

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-83507/10 Tool Kits, Electrical Connectors, Contacts and

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER Connector and Cable Accessories
FSC 5180

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

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